

```
import numpy as np
import pandas as pd
data = {
    "Article": [
        "cricket team won match",
        "football player scored goal",
        "cricket player hit century",

        "new smartphone launched",
        "artificial intelligence improves software",
        "new laptop has powerful processor",

        "government announced new policy",
        "election campaign started",
        "minister announced new scheme"
    ],

    "Category": [
        "Sports",
        "Sports",
        "Sports",

        "Technology",
        "Technology",
        "Technology",

        "Politics",
        "Politics",
        "Politics"
    ]
}
df = pd.DataFrame(data)
print("DATASET")
print(df)
categories = df["Category"].unique()
```

```

print("\nPRIOR PROBABILITY")
priors = {}
for category in categories:
    count = (df["Category"] == category).sum()
    prior = count / len(df)
    priors[category] = prior
    print(f"P({category}) = {count}/{len(df)} = {prior:.4f}")
word_counts = {}
total_words = {}
for category in categories:
    articles = df[df["Category"] == category]["Article"]
    words = []
    for article in articles:
        words.extend(article.lower().split())

    word_counts[category] = {}

    for word in words:
        word_counts[category][word] = word_counts[category].get(word, 0) + 1

    total_words[category] = len(words)

vocabulary = set()
for article in df["Article"]:
    vocabulary.update(article.lower().split())

vocab_size = len(vocabulary)

print("\nVocabulary Size:", vocab_size)

def word_probability(word, category):

    word = word.lower()

    count = word_counts[category].get(word, 0)

```

```

print("\nPRIOR PROBABILITY")
priors = {}
for category in categories:
    count = (df["Category"] == category).sum()
    prior = count / len(df)
    priors[category] = prior
    print(f"P({category}) = {count}/{len(df)} = {prior:.4f}")
word_counts = {}
total_words = {}
for category in categories:
    articles = df[df["Category"] == category]["Article"]
    words = []
    for article in articles:
        words.extend(article.lower().split())

    word_counts[category] = {}

    for word in words:
        word_counts[category][word] = word_counts[category].get(word, 0) + 1

    total_words[category] = len(words)

vocabulary = set()
for article in df["Article"]:
    vocabulary.update(article.lower().split())

vocab_size = len(vocabulary)

print("\nVocabulary Size:", vocab_size)

def word_probability(word, category):

    word = word.lower()

    count = word_counts[category].get(word, 0)

```

```

    probability = (count + 1) / (
        total_words[category] + vocab_size
    )

    return probability

def predict_article(article):

    words = article.lower().split()

    scores = {}

    print("\n" + "-" * 60)
    print("ARTICLE:", article)
    print("=" * 60)

    for category in categories:

        score = priors[category]

        print(f"\n{category}:")
        print(f"Prior = {priors[category]:.4f}")

        for word in words:

            probability = word_probability(word, category)

            print(
                f"P({word} | {category}) = {probability:.6f}"
            )

            score *= probability

        scores[category] = score

```

```
prediction = max(scores, key=scores.get)

print("\nFINAL SCORES")

for category, score in scores.items():
    print(f"{category} Score = {score:.10f}")

print("\nPredicted Category:", prediction)

return prediction
```

```
new_article = "cricket player won match"
```

```
predict_article(new_article)
```

```
predict_article("new smartphone software")
```

```
predict_article("government election policy")
```

```

DATASET
      Article      Category
0      cricket team won match      Sports
1      football player scored goal      Sports
2      cricket player hit century      Sports
3      new smartphone launched      Technology
4      artificial intelligence improves software      Technology
5      new laptop has powerful processor      Technology
6      government announced new policy      Politics
7      election campaign started      Politics
8      minister announced new scheme      Politics
    
```

```

PRIOR PROBABILITY
P(Sports) = 3/9 = 0.3333
P(Technology) = 3/9 = 0.3333
P(Politics) = 3/9 = 0.3333
    
```

Vocabulary Size: 29

```

=====
ARTICLE: cricket player won match
=====
    
```

```

Sports:
Prior = 0.3333
P(cricket | Sports) = 0.073171
P(player | Sports) = 0.073171
P(won | Sports) = 0.048780
P(match | Sports) = 0.048780
Score = 0.0000042465
    
```

```

Technology:
Prior = 0.3333
P(cricket | Technology) = 0.024390
P(player | Technology) = 0.024390
P(won | Technology) = 0.024390
P(match | Technology) = 0.024390
    
```

Score = 0.0000001180

Politics:

Prior = 0.3333

$P(\text{cricket} \mid \text{Politics}) = 0.025000$

$P(\text{player} \mid \text{Politics}) = 0.025000$

$P(\text{won} \mid \text{Politics}) = 0.025000$

$P(\text{match} \mid \text{Politics}) = 0.025000$

Score = 0.0000001302

FINAL SCORES

Sports Score = 0.0000042466

Technology Score = 0.0000001180

Politics Score = 0.0000001302

Predicted Category: Sports

=====
ARTICLE: new smartphone software
=====

Sports:

Prior = 0.3333

$P(\text{new} \mid \text{Sports}) = 0.024390$

$P(\text{smartphone} \mid \text{Sports}) = 0.024390$

$P(\text{software} \mid \text{Sports}) = 0.024390$

Score = 0.0000048365

Technology:

Prior = 0.3333

$P(\text{new} \mid \text{Technology}) = 0.073171$

$P(\text{smartphone} \mid \text{Technology}) = 0.048780$

$P(\text{software} \mid \text{Technology}) = 0.048780$

Score = 0.0000580375

Politics:

Prior = 0.3333

```
=====
ARTICLE: government election policy
=====
```

```
Sports:
Prior = 0.3333
P(government | Sports) = 0.024390
P(election | Sports) = 0.024390
P(policy | Sports) = 0.024390
Score = 0.0000048365
```

```
Technology:
Prior = 0.3333
P(government | Technology) = 0.024390
P(election | Technology) = 0.024390
P(policy | Technology) = 0.024390
Score = 0.0000048365
```

```
Politics:
Prior = 0.3333
P(government | Politics) = 0.050000
P(election | Politics) = 0.050000
P(policy | Politics) = 0.050000
Score = 0.0000416667
```

```
FINAL SCORES
Sports Score = 0.0000048365
Technology Score = 0.0000048365
Politics Score = 0.0000416667
```

```
Predicted Category: Politics
```

```
[1]: 'Politics'
```